

HYBRID QUANTUM TEMPORAL CONVOLUTIONAL NETWORKS

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ABSTRACT

Quantum machine learning models for sequential data face scalability challenges with complex multivariate signals. We introduce the Hybrid Quantum Temporal Convolutional Network (HQTCN), which combines classical temporal windowing with a quantum convolutional neural network core. By applying a shared quantum circuit across temporal windows, HQTCN captures long-range dependencies while achieving significant parameter reduction. Evaluated on synthetic NARMA sequences and high-dimensional EEG time-series, HQTCN performs competitively with classical baselines on univariate data and outperforms all baselines on multivariate tasks. The model demonstrates particular strength under data-limited conditions, maintaining high performance with substantially fewer parameters than conventional approaches. These results establish HQTCN as a parameter-efficient approach for multivariate time-series analysis.

Index Terms— Quantum Machine Learning, Multivariate Time-series, Quantum Temporal Convolutional Networks

1. INTRODUCTION

Sequential data analysis presents significant challenges across many domains. While classical models have achieved considerable success, they often struggle to efficiently capture complex, long-range dependencies in high-dimensional time-series. Quantum machine learning (QML) offers a promising approach to address these limitations. Through quantum effects such as superposition and entanglement, QML models can process information in exponentially large Hilbert spaces [1], potentially enabling more efficient learning of complex correlations in sequential data. Additionally, QML models have shown promise for data-efficient learning with relatively few parameters [2].

Recent QML research has explored quantum adaptations of classical sequential models. Quantum Long Short-Term

Memory (QLSTM) [3] and Quantum Recurrent Neural Networks (QRNN) [4, 5] demonstrated how variational quantum circuits (VQCs) could process temporal data. However, recurrent quantum architectures face inherent limitations: their sequential processing creates bottlenecks for learning long-range dependencies, backpropagation-through-time (BPTT) requires significant computing resources and quantum mechanical constraints complicate the propagation of quantum states across time steps, limiting scalability on near-term quantum devices.

We propose the Hybrid Quantum Temporal Convolutional Network (HQTCN), which combines classical fully-connected layers with quantum convolutional neural network (QCNN) components. Inspired by classical temporal convolutional networks that use dilated convolutions and residual connections for efficient long-range modeling, our approach employs a temporal sliding window strategy that applies dilated sampling to input sequences before processing them through shared quantum circuits. This temporal windowing mechanism enables the quantum components to effectively capture dependencies across different time scales while significantly reducing parameter overhead, making it suitable for scalable processing of high-dimensional multivariate time-series.

2. RELATED WORKS

2.1. Recurrent Quantum Models

Early quantum approaches to sequential data processing adapted classical recurrent architectures, developing QRNNs and QLSTMs [3, 4, 5]. These models use VQCs as recurrent cells to process sequence elements iteratively, showing promise in simulation studies for forecasting tasks [6, 7].

However, recurrent quantum models face both architectural and fundamental limitations. Architecturally, their sequential processing creates bottlenecks when learning long-range dependencies in complex multivariate time-series. More critically, quantum mechanical constraints pose implementation challenges: the no-cloning theorem prevents perfect copying of quantum states, while accessing information from quantum hidden states requires measurements that

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collapse superpositions and destroy quantum information [8].

Practical implementations must rely on mid-circuit measurements and classical feed-forward operations, which introduce noise and computational overhead on current Noisy Intermediate-Scale Quantum (NISQ) devices [9]. These limitations motivate exploring novel quantum architectures that can better leverage quantum advantages for time-series analysis.

2.2. Classical Temporal Convolutional Networks (TCNs)

TCNs have emerged as an effective alternative to RNNs for sequence modeling [10]. TCNs employ causal convolutions that respect temporal ordering and dilated convolutions that expand the receptive field by sampling inputs with increasing stride patterns. By stacking residual blocks with exponentially increasing dilation factors, TCNs efficiently capture long-range dependencies while enabling parallel processing during training.

Despite these advantages, classical TCNs may require substantial depth and parameters to model complex, high-dimensional time-series with intricate correlations. This computational overhead suggests potential benefits from quantum enhancement, where quantum superposition and entanglement could provide more efficient representations of complex temporal patterns.

2.3. Quantum Convolutional Neural Network (QCNN)

QCNNs extend classical CNNs to the quantum domain, offering advantages in parameter efficiency and noise resilience [11]. QCNNs employ a hierarchical structure alternating between convolutional and pooling layers implemented as quantum operations (Figure 1(a)).

The QCNN processes input data x by encoding it into an n -qubit quantum state $\rho(x)$, then applying L layers of quantum operations. Each convolutional layer applies parameterized unitary transformations $\mathcal{U}^{(\ell)}(\theta)$ to overlapping qubits:

$$\rho \mapsto \mathcal{U}^{(\ell)\dagger}(\theta) \rho \mathcal{U}^{(\ell)}(\theta) \quad (1)$$

To mimic a classical pooling layer, which reduces input dimensions, a partial trace operation $\mathcal{P}_B$ is applied to trace out a subset of qubits B . Specifically, let $A = \{1, \dots, n\}$ be the qubits (indices) of the entire system and a sub-system $B \subseteq A$, the trace operator $\mathcal{P}_B$ acting on density state ρ in $\mathcal{H}_A$ is defined via:

$$\langle v | \mathcal{P}_B(\rho) | w \rangle_B := \sum_j \langle v \otimes e_j | \rho | w \otimes e_j \rangle_A \quad (2)$$

where v, w are arbitrary quantum states in Hilbert space $\mathcal{H}_{A \setminus B}$ and e_i is any basis in $\mathcal{H}_B$. Together, the actions of

$\mathcal{P}^{(\ell)}$ and $\mathcal{U}^{(\ell)}$ are combined as:

$$\begin{aligned} \langle v | \mathcal{P}_B \left(\mathcal{U}^{(\ell)\dagger}(\theta) \rho \mathcal{U}^{(\ell)}(\theta) \right) | w \rangle_B := \\ \sum_j \langle v \otimes e_j | \mathcal{U}^{(\ell)\dagger}(\theta) \rho \mathcal{U}^{(\ell)}(\theta) | w \otimes e_j \rangle_A \end{aligned} \quad (3)$$

The joint action of $\mathcal{P}^{(\ell)}$ and $\mathcal{U}^{(\ell)}$ is defined as an operator $\mathcal{C}^{(\ell)}$ via Eq. (3).

Starting with an initial state $\rho^{(0)}$, the state is transformed layer by layer:

$$\rho^{(\ell)} = \mathcal{C}^{(\ell)}(\rho^{(\ell-1)}) \quad \text{for } \ell = 1, 2, \dots, L \quad (4)$$

The final output of the network is the expectation value of a Hermitian observable H measured on the final state $\rho_\theta^{(L)}$:

$$f_\theta(x) = \text{Tr}[H \rho_\theta^{(L)}(x)]. \quad (5)$$

where the function bears the parameters θ due to the parameterized layers involved in Eq. (1).

QCNNs offer two key advantages for near-term quantum devices: parameter efficiency with only $O(\log(n))$ variational parameters for n -qubits [11], and provable barren plateau resilience that ensures trainability as the system size increases [12]. These properties make QCNN a strong candidate for building a quantum analogue of the TCN.

3. HYBRID QUANTUM TEMPORAL CONVOLUTIONAL NETWORKS

The HQTCN architecture explores a deeper integration of quantum processing by using a quantum circuit as the core computational unit in a recurrent, temporal window fashion. Its architecture, shown in Figure 1(b), functions analogously to an RNN.

Let $X \in \mathbb{R}^{C \times T}$ denote a multivariate time series with C channels and T time steps indexed by $t = 0, 1, \dots, T-1$. A sliding window is applied along the temporal dimension: at each valid step t , a receptive field $W_t \in \mathbb{R}^{C \times K}$ is formed by sampling K points with dilation $d \in \mathbb{N}$ at indices

$$\{t - d(K-1), \dots, t - d, t\}.$$

To remain within range, the window is applied only for $d(K-1) \leq t \leq T-1$.

Each window W_t is flattened into $\mathbf{w}_t \in \mathbb{R}^{CK}$, which is typically too high-dimensional for direct quantum input. A fully-connected embedding layer with weight matrix $W_{fc} \in \mathbb{R}^{CK \times n}$ and bias $\mathbf{b}_{fc} \in \mathbb{R}^n$ maps this vector to

$$\mathbf{e}_t = \mathbf{w}_t W_{fc} + \mathbf{b}_{fc}, \quad (6)$$

yielding an embedding $\mathbf{e}_t \in \mathbb{R}^n$ that matches the number of qubits n in the quantum circuit.

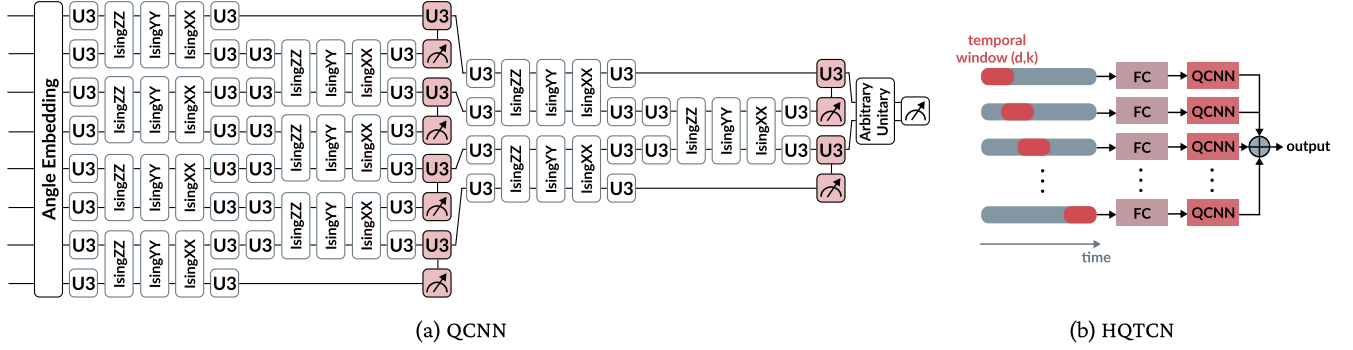


Fig. 1. Model Architectures. (a) A standalone QCNN with hierarchical quantum convolutional and pooling layers. (b) The proposed HQTCN, which applies a shared QCNN circuit over a temporal sliding window before aggregating the outputs.

The core of the model is a shared QCNN circuit f_{θ} of Eq. (5), parametrized by the trainable parameters $\theta = \{\theta_{conv}, \theta_{pool}\}$. Each step t , the circuit processes embedding $\mathbf{e}_t$ to produce:

$$o_t = f_{\theta}(\mathbf{e}_t). \quad (7)$$

Recurrent application of the quantum circuit (cell) from $t = d(K-1)$ to $T-1$ yields sequence $(o_{d(K-1)}, \dots, o_{T-1})$.

A single prediction $\hat{Y}$ for the entire input is then obtained by aggregating these outputs, for example by averaging:

$$\hat{Y} = \frac{1}{T - d(K-1)} \sum_{t=d(K-1)}^{T-1} o_t. \quad (8)$$

This recurrent structure captures temporal dependencies by consistently applying the same quantum processing cell across the sequence.

4. EXPERIMENT

4.1. Data

4.1.1. NARMA

We evaluate time-series prediction using the Non-linear Autoregressive Moving Average (NARMA) dataset [13], a standard benchmark for sequential models [14, 15]. The 10th-order NARMA sequence is generated by:

$$y_t = 0.3y_{t-1} + 0.05y_{t-1} \sum_{i=0}^9 y_{t-i-1} + 1.5u_{t-10}u_{t-1} + 0.1 \quad (9)$$

where u is an i.i.d. random input sequence. This task tests models' ability to capture complex long-term dependencies. We generated 240 time points, split into 70% training, 15% validation, and 15% testing.

4.1.2. PhysioNet EEG

We use the motor-imagery subset from PhysioNet EEG [16] recorded with BCI2000 [17]. From 109 subjects, we selected

left- versus right-hand imagery data from 50 randomly chosen subjects. The data was sampled at 80 Hz across 64 scalp channels for 249 time steps for the binary hand movement classification task.

4.2. Experimental Setup

We compare HQTCN against classical TCN [10], LSTM [18], Transformer [19], and QLSTM [3]. QLSTM was excluded from PhysioNet experiments due to its limitations with multi-variate data. We include a standalone QCNN [11] for ablation analysis of our temporal window approach.

Models were implemented using PennyLane [20] and PyTorch on a Linux server with 128 CPU cores (256 threads), 503 GB of RAM, and an NVIDIA A100-PCIE GPU with 40 GB of memory. The software stack consisted of Python 3.11.7, PyTorch 2.5.0+cu121, and CUDA 12.1. All QCNN-based models used 8-qubits with two layers (convolution + pooling each).

Training hyperparameters were: NARMA (kernel size 5, dilation 2, learning rate 0.005, weight decay 0.0001); PhysioNet EEG (kernel size 12, dilation 3, learning rate 0.001, weight decay 0.0001).

5. RESULTS

Table 1. NARMA Dataset Results. Comparison of mean test loss ($\pm$ std. dev. over three seeds) and the number of trainable parameters for all models on the NARMA prediction task.

	Test Loss	# of Parameters
LSTM	0.0387 ± 0.0111	531
Transformer	0.0423 ± 0.0127	8,641
TCN	0.0380 ± 0.0032	10,785
QLSTM	0.0547 ± 0.0192	126
QCNN	0.0456 ± 0.0181	2,192
HQTCN	0.0487 ± 0.0193	312

Table 2. PhysioNet EEG Analysis Results. Test AUROC ($\pm$ std. dev. over three seeds) and parameter counts for all models on the EEG classification task across performance-optimized, parameter-constrained, and sample-constrained (N=10) settings.

	Performance-optimized		Parameter-constrained		Sample-constrained (N = 10)
	Test AUROC	# of Parameters	Test AUROC	# of Parameters	Test AUROC
LSTM	0.7552 ± 0.0182	66,625	0.7657 ± 0.0178	5,265	0.6582 ± 0.0832
Transformer	0.7672 ± 0.0269	19,201	0.7618 ± 0.0261	7,729	0.6651 ± 0.0645
TCN	0.7815 ± 0.0296	49,985	0.7755 ± 0.0245	5,857	0.5875 ± 0.0528
QCNN	0.6649 ± 0.0342	127,760	0.6649 ± 0.0342	127,760	0.5112 ± 0.0072
HQTCN	0.7929 ± 0.0383	6,416	0.7929 ± 0.0383	6,416	0.6713 ± 0.0515

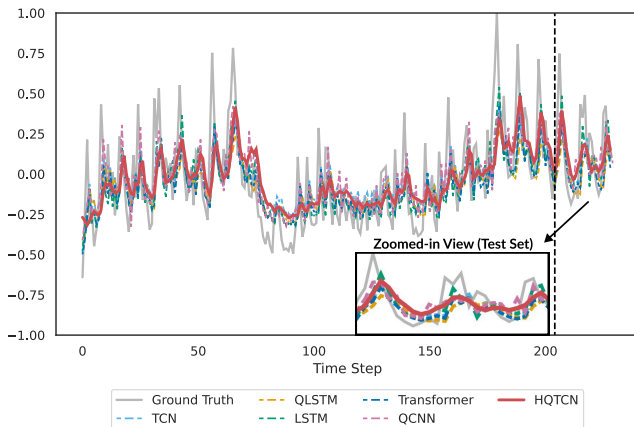


Fig. 2. NARMA Time-Series Prediction Results. Comparison of HQTCN’s predictions (red) against the ground truth (gray) and other models on the NARMA test set. The zoomed-in view highlights the test-set performance.

On the NARMA dataset (Table 1, Fig. 2), HQTCN achieves competitive performance with a test loss of 0.0487 ± 0.0193 , comparable to classical TCN (0.0380 ± 0.0032) and QLSTM (0.0547 ± 0.0192).

HQTCN demonstrates superior performance on the multivariate PhysioNet EEG dataset (Table 2). In the performance-optimized setting, where all baseline models were trained with their optimal number of parameters to maximize performance, HQTCN achieves the highest test AUROC of 0.7929 ± 0.0383 despite having significantly fewer parameters (6,416 total: 6,152 classical + 264 quantum) than classical baselines.

This advantage is further validated in the parameter-constrained scenario, where classical baseline models’ parameters were reduced to approximately match HQTCN’s parameter count. HQTCN maintained superior performance against these parameter-matched baselines, demonstrating exceptional parameter efficiency.

The model’s efficiency is particularly evident in data-limited scenarios. With only 10 subjects, HQTCN achieves an AUROC of 0.6713 ± 0.0515 , significantly outperforming classical TCN (0.5875 ± 0.0528) and QCNN (0.5112 ± 0.0072).

Our temporal sliding window approach addresses a key

limitation of standalone QCNN: since QCNNs are not designed for multivariate temporal data, they require extensive classical preprocessing layers that scale with input dimensionality. Our windowing architecture circumvents this issue, enabling HQTCN to process multivariate sequences with $20\times$ fewer parameters than standalone QCNN (6,416 vs 127,760) while improving generalizability. This efficiency makes HQTCN particularly effective for high-dimensional, multivariate time-series such as EEG signals, where it significantly outperforms baselines.

6. CONCLUSION

We introduced HQTCN, a novel hybrid quantum-classical model for multivariate time-series analysis. While achieving comparable performance to baselines on univariate NARMA data, HQTCN significantly outperformed all classical and quantum models on the high-dimensional EEG dataset.

The model’s architecture proves critical to its success. Ablation studies confirm that integrating classical temporal sliding windows with quantum convolutional components is essential for effective temporal sequence processing, with HQTCN substantially outperforming standalone QCNN models.

HQTCN achieves superior performance with exceptional parameter efficiency, using from 3- $20\times$ fewer parameters than baseline models while maintaining competitive accuracy. The model also demonstrates strong sample efficiency, outperforming all baseline models when trained on limited data (10 subjects).

We attribute this high efficiency to the inherent properties of quantum computing. By leveraging superposition and entanglement, the quantum components of our model can access a vast feature space, enhancing the model’s expressibility and allowing it to learn key patterns and generalize effectively from a limited number of parameters and samples [1, 2, 21].

These results suggest that HQTCN represents a promising approach toward practical quantum advantage in complex time-series analysis, particularly for high-dimensional multivariate data where parameter efficiency and sample efficiency are crucial.

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